

MATH 3260 - Chapter 7 Worksheets

Eigenvalues and Eigenvectors

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 7.1 - Eigenvalues and Eigenvectors

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Verify that $(1, 1)$ and $(1, -1)$ are eigenvectors of

$$\begin{bmatrix} 20 & 1 \\ 1 & 20 \end{bmatrix}$$

and state their eigenvalues. Explain why the zero vector is not allowed to be an eigenvector.

2. Find the characteristic polynomial and eigenvalues of

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

3. For each eigenvalue in Problem 2, explicitly form $A - \lambda I$, solve

$$(A - \lambda I)\mathbf{v} = \mathbf{0},$$

and give a basis for the eigenspace.

4. For

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix},$$

find the algebraic and geometric multiplicity of $\lambda = 2$. Explain why the two multiplicities are different.

5. Prove directly that $E_\lambda = \ker(A - \lambda I)$ is closed under addition.

6. Find the eigenvalues of the triangular matrix

$$\begin{bmatrix} 9 & 2 & -1 \\ 0 & 20 & 4 \\ 0 & 0 & 21 \end{bmatrix}.$$

7. For $T(p) = p + xp'$ on P_2 , find $T(2 - 3x + 4x^2)$ and identify an eigenbasis for T .

Section 7.2 - Diagonalization

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Prove that similar matrices have the same determinant.

2. Diagonalize

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix}$$

using the supplied eigenvectors

$$\mathbf{v}_1 = (-1, 0, 1)^T, \quad \mathbf{v}_2 = (-1, 1, 1)^T, \quad \mathbf{v}_3 = (-1, 1, 0)^T$$

with eigenvalues 1, 2, 3, respectively. Construct P and D .

3. Verify directly that $AP = PD$ for the matrices in Problem 2.

4. Use

$$A = PDP^{-1}$$

to compute

$$A^4 = PD^4P^{-1}.$$

5. Determine whether

$$J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

is diagonalizable. Justify using eigenspace dimension.

6. Prove that eigenvectors associated with two distinct eigenvalues are linearly independent.

7. Explain why a 3×3 matrix with three distinct eigenvalues must be diagonalizable.

Section 7.3 - Symmetric Matrices and Orthogonal Diagonalization

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Determine whether

$$\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$$

is symmetric.

2. Prove that if the columns of P are orthonormal, then $P^{-1} = P^T$.

3. Orthogonally diagonalize

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

Show the normalization of the eigenvectors.

4. Verify directly that the eigenvectors $(2, 1)$ and $(-1, 2)$ are orthogonal.

5. For

$$C = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix},$$

verify that $(1, 1, 1)$ is an eigenvector and state its eigenvalue.

6. Apply one Gram-Schmidt step to $(1, -1, 0)$ and $(1, 0, -1)$.

7. Prove that an orthogonally diagonalizable real matrix is symmetric.

Section 7.4 - Applications of Eigenvalues and Eigenvectors

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation. Use the following Leslie matrix for Problems 1-2:

$$L = \begin{bmatrix} 0 & \frac{6}{5} & 2 \\ \frac{11}{20} & 0 & 0 \\ 0 & \frac{11}{20} & 0 \end{bmatrix}.$$

1. Verify that $(20, 10, 5)^T$ is an eigenvector of L and find its eigenvalue.
2. Convert the stable age vector in Problem 1 to percentages that sum to 100%.

3. Solve

$$\mathbf{y}' = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \mathbf{y}, \quad \mathbf{y}(0) = (3, 1)^T.$$

Use eigenvalues and eigenvectors.

4. Find the symmetric matrix A associated with

$$q(x, y) = 5x^2 + 4xy + 2y^2.$$

5. Use

$$P = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}, \quad P^T A P = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix},$$

to transform

$$5x^2 + 4xy + 2y^2 = 12$$

into principal-axis coordinates. Explain why the cross term disappears.

6. Classify $q(x, y) = x^2 - 4y^2$ using the signs of its eigenvalues.

7. For

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix},$$

use the Rayleigh quotient

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$

and the eigenvalues of A to find the maximum and minimum of $5x^2 + 4xy + 2y^2$ subject to $x^2 + y^2 = 1$.

8. Prove that an orthogonal change of coordinates preserves vector length.